3 Set Theory

We cover Section 3.1-3.4 of the text, however have rearranged material so our first three sections don't agree exactly with the text. We also include material from Section 5.5 of the text in our Section 3.3.

Topics Covered:

- Set-builder notation, subset, emptyset
- Set operations, set laws, Venn diagrams
- Cardinality, power-set, inclusion-exclusion
- The Pigeonhole principle, counting by bijections
- Probability

3.1 Sets and Subsets

3.1.1 Basic Definitions

The notation of symbolic logic allows us to give precise definitions for many notions we have until now defined only intuitively.

We still leave the idea of a set A a bit imprecise: 'a collection of distinct elements from a universe', but given this, we lets look at some related definitions.

Given two sets A and B with the same universe U, A *is a subset of* B, written $A \subseteq B$, if

$$\forall x[x \in A \to x \in B]$$

.

Problem 3.1. Given this definition, is $A \subseteq A$. Where $\emptyset = \{\ \}$ is *the empty set*, is $\emptyset \subseteq A$? Is $\{3, 1, 2, 2\} \subseteq \{1, 2, 3\}$.

How would you define the statement $A = B$, that the sets A and B are the same set? One way would be

$$\forall x[x \in A \iff x \in B]$$

.

But look at the following logical equivalence:

$$\begin{aligned} \forall x[x \in A \iff x \in B] &\iff \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in A] \\ &\iff A \subseteq B \wedge B \subseteq A \end{aligned}$$

This rigourously proves the obvious statement that

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

As simple as this seems, we will use it a lot in proofs.

A set A is a *proper subset* of B, written $A \subset B$ if $A \subseteq B$ and $A \neq B$.

Give and alternate definition of $A \subset B$:

$A \subseteq B$ and _____.

As the basic set definitions have 'quantified open statment' versions, proofs about set theory will be very similar to proofs of logical equivalences and logical implications. Let's see one.

Theorem 3.1. *Let $A, B, C \subseteq U$.*

i. If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

ii. If $A \subset B$ and $B \subseteq C$ then $A \subset C$.

iii. If $A \subseteq B$ and $B \subset C$ then $A \subset C$.

iv. If $A \subset B$ and $B \subset C$ then $A \subset C$.

Proof. We prove only *i* and *iii*. To prove *i*, assume the premises, that $A \subseteq B$ and $B \subseteq C$. By definition this means that

$$\begin{aligned} & \forall x[x \in A \to x \in B] \wedge \forall x[x \in B \to x \in C] \\ \iff & \forall x[(x \in A \to x \in B) \wedge (x \in B \to x \in C)] \\ \Rightarrow & \forall x[(x \in A \to x \in C] \text{ by syllogism.} \end{aligned}$$

This last statement is the definition of $A \subseteq B$.

Now we prove *iii*. Assume that $A \subseteq B$ and $B \subset C$. As $B \subset C$ implies $B \subseteq C$ we have the premises of part *iii.*, so we may conclude that $A \subseteq C$. Thus we only have to show that there is some x that is in C but not A. By the premise $B \subset C$ there is some x in C that is not in B. As A is a subset of B, this x is not in A either. So we are done. □

Notice that our proof style differed from part *i* to part *c*. In *a* we were very notational while in *c* we relied a lot more on english sentences. Which one is easier for you to read? Mathematics tends to be written as in part *c*, as it is easier to read. But it is still precise, many words or phrases have more restricted, or different usage in Math than they do in regular conversation. It is good to learn some of these usages.

Note on English

The word 'assume' is usually comes at the beginning of a proof and will be followed, often much later, by

- a conclusion, if we are proving an argument to be valid; or
- a contradiction, when we are doing proof by contradiction.

When the proofs involved between the assumptions and the conclusion/contradiction are very short, we often use 'if' instead of 'assume'. Ex. 'If x is in A then it is also in B, which is false, so x is not in A.'

Problem 3.2. Finish the proof of the Theorem. Try to write it more like the proof of part *iii* above.

3.1.2 Set-builder Notation

As for these definitions, universal quantification is usefel in describing specific sets. This is seen With *set-builder* notation. Given a universe U and an open statement $p(x)$,

$$\{x \in U \mid p(x)\}$$

(or simply $\{x \mid p(x)\}$ when the universe is understood), is the set of elements in U for which p is true.

For example the set $2\mathbb{N} = \{0, 2, 4, 6, \dots\}$ can be described by $2\mathbb{N} = \{x \in \mathbb{Z} \mid x \geq 0, 2 \text{ divides } x\}$.

As a variation that works when the universe is understood, one can even use a function of x: $2\mathbb{N} = \{2x \mid x \geq 0\}$.

3.2 Set Operations and Laws of Set Theory

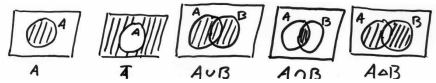
Each of the basic logical connectives defines an operation on sets.

Let A and B be sets with the same universe U, then we have

- the *complement* $\overline{A} = \{x \mid x \notin A\}$
- the *union* $A \cup B = \{x \mid x \in A \vee x \in B\}$

- the *intersection* $A \cap B = \{x \mid x \in A \wedge x \in B\}$, and
- the *symmetric difference* $A \triangle B = \{x \mid x \in A \underline{\vee} x \in B\}$.

These set operations can be visualised by a *Venn (or circle) Diagram.*



There is also the *relative complement of A in B* defined as

$$A - B = \{x \mid x \in A \vee x \notin B\}.$$

Problem 3.3. Prove that $A - B = A \cap \overline{B}$

Problem 3.4. What is the set $\{x \mid x \in A \to x \in B\}$? Perhaps use the implication law. Draw a Venn diagram to describe it. How does this relate to $A - B$.

As the sets operations are defined from logical connectives, it makes sense that they exhibit similar properties, such as DeMorgan's Law:

$$\overline{A \cup B} = \overline{A} \cap \overline{B}$$

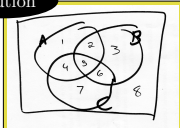
The proof of this follows directly from DeMorgans Law for logic:

$$\begin{aligned}
\overline{A \cup B} &= \overline{\{x \mid x \in A \vee x \in B\}} && \text{def of union} \\
&= \{x \mid \neg(x \in A \vee x \in B)\} && \text{def of complement} \\
&= \{x \mid x \notin A \wedge x \notin B\} && \text{DeMorgan} \\
&= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} && \text{def of complement} \\
&= \overline{A} \cap \overline{B}
\end{aligned}$$

Similarily, we get the Set Laws, set theoretic versions of all of the laws of logic. See the table on page 139 of the text. It is useful to also include the identity from Problem 3.3, which we view as an analogue of the implication law, in this list.

Example 3.2. Show that $A - (B \cup C) = (A - B) \cap (A - C)$.

Solution



One way to do this is to use a Venn diagram. One can check that both sides of the equation are region 1 in the figure on the left. Alternately we can use the set laws.

$$\begin{aligned} A-(B\cup C) &= A\cap\overline{B\cup C} && \text{Implication Law}\\ &= A\cap\overline{B}\cap\overline{C} && \text{DeMorgan}\\ &= A\cap A\cap\overline{B}\cap\overline{C} && \text{Idempotence}\\ &= (A\cap\overline{B})\cap(A\cap\overline{C}) && \text{Commutativity}\\ &= (A-B)\cap(A-C) && \text{Implication}\end{aligned}$$

3.3 More on counting

3.3.1 Double-Counting

The *cardinality* $|A|$ of a finite set A is the number of elements in the set. For infinite sets such as $\mathbb{Z}$ or $\mathbb{R}$, we say the cardinality is ∞. In other classes you will see that there are various infinite cardinalities, but that won't interest us in this class.

The power set of $\mathscr{P}(A)$ of a set A is the set of all subsets of A. If $|A| = n$, what is $|\mathscr{P}(A)|$?

One easy way to count it is using the product rule. In choosing an element of $\mathscr{P}(A)$, that is, a subset S of A, we have to choose for each of the n elements x of A whether or not x is in S. There are 2^n ways to do this, so $\mathscr{P}(A) = 2^n$.

Another way to count is to use the sum rule. Counting the 0 element sets, then the 1 element sets, et cetera, we get $|\mathscr{P}(A)| = \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n}$. Thus

$$2^n = \sum_{i=0}^{n} \binom{n}{i}.$$

The proof we have given of this identity is an example of what is called *double counting*. We counted a set in two different ways to arrive at the identity. This is a common technique for proving combinatorial indentities.

Problem 3.5. Show that this identity also follows from the Binomial theorem.

Problem 3.6. There are n (distinct) cats and 1 dog signed up for a commitee that will have r members. Count the number of ways of to select the commitee members, in two different ways, to show that

$$\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$$

3.3.2 Counting by Bijections

We are good at counting permutations and combinations of various types. It makes sense to relate other counting problems to these problems. We saw this when counting the number of positive integer solutions to the equation $x_1 + x_2 + x_3 = 10$, but it is a fundamental idea.

Given two sets A and B, if we can define a *bijection* (a 1-to-1, onto function) from A to B. Then $|A| = |B|$.

Even when we know the cardinality of a set B it can be useful to get a bijection with another well know set A.

Example 3.3. We define a bijection f from the powerset $\mathscr{P}([n]$ to the set of n digit binary strings by letting the ith digit of $S \in \mathscr{P}([n])$ be 1 if $i \in S$ and 0 otherwise.

We can verify that it is a bijection by exhibiting an *inverse* function g from the set of n-digit binary strings to $\mathscr{P}([n])$: $g(b_n b_{n-1} \ldots b_2 b_1)$ contains the element i if and only if b_i is 1.

Why is this useful, now we have an *ordering* of $\mathscr{P}([n])$ by converting these strings to the integers they represent. If we want to write an algorithm checking some property of subsets of $[n]$, this is useful.

A *composition* of the number n is an expression of n as a sum of positive integers. So $5 = 1+1+1+1+1$, $5 = 1+2+1+1$, $5 = 2+1+1+1$, $5 = 4+1$. How many compositions are there of the number n?

We could count the number of compositons with i summands for each i, and then add them up. (See Section 1.4 of the text.) But we will count the set by bijections.

Indeed, we will show that there are 2^{n-1} compositions of n by giving a bijection between the set of compositions, and $\mathscr{P}([n-1]$.

Start with the string $(1+1+\cdots+1+1)$. There are $n-1$ occurences of the symbol '+'. For a set $S \in \mathscr{P}([n-1])$, let $f(S)$ be the composition constructed as follows. If i is in S replace the ith occurence of '+' with ')+('. So the subset $1, 3, 4 \subset [8]$ corresponds to the string

$$(1) + (1+1) + (1) + (1+1+1+1).$$

Then collapse the bracketed sums: $1+2+1+4$.

One can check that this is a bijection.

Problem 3.7. Using the ordering of $\mathscr{P}([7])$ you get from the bijection of it to the 7-digit binary strings, give the first three compositions of [8].

3.3.3 k - to - 1 functions

Generalising the idea of counting by bijections or 1-to-1 functions, we can count by k-to-1 functions.

Given two sets A and B, if we can define a function $f : A \to B$ such that for each b in B, there are k elements a of A with $f(a) = b$, then $|A| = k|B|$.

We saw this in counting the number of combinations of a set, though we didn't explicitly talk about a function.

To count the number of k-combinations of n we define a function

$$f : (x_1, x_2, \ldots, x_k) \mapsto \{x_1, x_2, \ldots, x_k\}$$

from the set of k-permutations to the set of k-combinations. For each k-combination C, there are $k!$ different k-permutations σ with $f(\sigma) = C$. So f is $k!$-to-1.

We concluded that the number $C(n, k)$ of k-combinations was

$$C(n, k) = \frac{P(n, k)}{k!} = \frac{n!}{k! \cdots (n-k)!}.$$

3.3.4 The Pigeonhole Principle

A variation on the idea of counting by bijections it the idea that if we can define an onto function f from A to B, that is not a bijection, then $|A| > |B|$.

It is usually stated in the converse, with the following example:

> If there are $n + 1$ pigeons in n pigeonholes, then some hole contains more than one pigeon.

This is a fairly obvious example, but the pigeonhole priciple can be used in more complicated problems.

Problem 3.8. Show that for any 5 points in the equilateral triangle of side length 1, there must be two whose distance apart is at most 1/2.

Example 3.4. A golfer plays 132 games over 11 weeks, playing at least one game a day. Show that there is some stretch of consecutive days in which she plays exactly 21 games.

Solution

Let a_i be the number of games she plays by the end of the ith day. The 154 numbers $a_1, a_2, \ldots, a_{121}$, and $b_1 = a_1 + 21, b_2 = a_2 + 21, \ldots, b_{121} = a_{121} + 21$ are all between 1 and 153, so some two of them are the same. By the condition that she plays at least 1 game a day, all of the a_i are distinct, and all of the b_i are distinct, so there are some i and j such that $b_i = a_j$. So the golfer plays 21 games from the $i + 1^{th}$ to the jth day.

Problem 3.9. At a party, any two people are either acquainted, or are strangers. How many people must there be at the party so that there are either some three people that are mutually acquainted, or there are some three people that are mutual strangers.

3.3.5 Inclusion and Exclusion

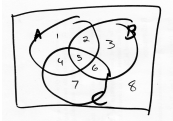
Observe from the Venn diagram that $|A \cup B| = |A| + |B| - |A \cap B|$. We can use this to count the number of integers in [60] that are multiples of 2 or 3:

Solution

Let A be the set of multiples of 2 and B be the set of multiples of 3. Then $|A| = 30$ and $|B| = 20$ and $A \cap B$ is the set of multiples of 6, so $|A \cap B| = 10$. So by the above identity we have that

$$|A \cup B| = |A| + |B| - |A \cap B| = 30 + 20 - 10 = 40$$

Problem 3.10. How many arrangements of the [6] start or end with an odd digit?



What about for more circles in the Venn Diagram? We see that $|A \cup B \cup C|$ is equal to

$$\begin{aligned} & |A| + |B| + |C| \\ - \quad & |A \cap B| - |A \cap C| - |B \cap C| \\ + \quad & |A \cap B \cap C| \end{aligned}$$

Example 3.5. Let's count the number of arrangements of the letters of the word CHEMIST that have H before T, or T before E, or E before M.

Let A_1 be the set of arrangements with H before T. Let A_2 be the set of arrangements with T before E. Let A_3 be the set of arrangements with E before M.

Then $|A_1| = \frac{7!}{2!} = |A_2| = |A_3|$, $|A_1 \cap A_2| = \frac{7!}{3!} = |A_2 \cap A_3|$, $|A_1 \cap A_3| = \frac{7!}{2!2!}$, and $|A_1 \cap A_2 \cap A_3| = \frac{7!}{4!}$.

Using the above identity, we can put this together to get $|A_1 \cap A_2 \cap A_3|$.

3.4 A First Word on Probability

Probability is a convenient way of counting the relative sizes of sets.

Definition 3.6. An *experiment* is an activity with some set of possible outcomes. The *sample space* is the set S of possible outcomes. An *event* is a subset A of the sample space.

Example 3.7. In an experiment we toss two coins. Each coin can come up 'Heads' (H) or 'Tails' (T), so our sample space is $S = \{HH, HT, TH, TT\}$. Some possibles events are

- $A_1 = \emptyset$.
- $A_2 = \{HT, TH\}$.
- $A_3 = S$.

If an experiment has a sample space in which each outcome is equally likely, the probability $Pr(A)$ of an event A is

$$Pr(A) = |A|/|S|.$$

We write $Pr(a)$ for $Pr(\{a\})$.

Example 3.8. Continuing the previous example, we have that $Pr(A_1) = 0$, $Pr(A_2) = 1/2$, and $Pr(A_3) = 1$.

Example 3.9. Two distinguishable dice are rolled. Find the probability that the rolled numbers sum to a) 7, b) 9, c) 12. d) Find the probability that at least one 2 is rolled.

Solution

The sample space S has 36 elements.
a) $Pr(\{16, 25, 34, 43, 52, 61\}) = \frac{6}{36} = 1/6$
b) $Pr(\{36, 45, 54, 63\}) = \frac{4}{36} = 1/9$
c) $Pr(\{66\}) = 1/36$
d) Let A_1 be the event that the first die is a 2 and let A_2 be the event that the second die is a 2. The we are looking for

$$\begin{aligned} Pr(A_1 \cup A_2) &= \frac{1}{36}|A_1 \cup A_2| \\ &= \frac{1}{36}(|A_1| + |A_2| - |A_1 \cap A_2|) \\ &= \frac{1}{36}(6 + 6 - 1) = \frac{11}{36}. \end{aligned}$$

Problem 3.11. You choose 2 Jellybellies from a cup of 30 different Jellybellies. What is the probability of choosing

a) banana and lemon

b) neither banana or lemon

c) banana but not lemon

Problem 3.12. In a group of 80 students

- 45 take Math
- 48 take CompSci
- 38 take Stats
- 30 take Math and CompSci
- 30 take CompSci and Stats
- 23 take Math and Stats
- 15 take all three

a) How many students take none of Math CompSci or Stats?

b) Find the probability that a randomly chosen student takes Math.

c) Find the probability that a randomly chosen Stats student also takes CompSci.